

Pulsed, Not Pressed: State-Space Injection Matches the Judged Surprise of Text Interruption in Open-Ended Generation

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Abstract

A research program on machine novelty has intervened at three depths of a language model — the token distribution, the context, and the weights — and measured the limit of each. This paper opens the fourth: the residual stream during live generation. We introduce a rank-1/2 operator algebra on hidden states (translations toward model-derived concept directions, closed-form rotors, projections, repulsion from a moving mean; an anchor and a norm shell as guards; doses in units of the premise-state norm) and evaluate it on Qwen3-8B-Base under a deception-resistant protocol: a blind judge scores surprise, connection and coherence (0–10) on windows of generated-only text, paired over premises, every confirmatory analysis pre-registered. Four pre-registered findings and one descriptive. (i) Continuous injection has no fertile band — more dose, less judged surprise, on both carriers tested — while the same directions pulsed for 32 tokens every 300 on a habituated carrier raise it and climb a dose ladder where the continuous regime degenerates or is inert (the head-to-head at equal dose is a trend at ten premises). (ii) Guards change what a dose does rather than admitting a larger one; at guarded dose 2.0 the pulse is equivalent to text interruption in judged surprise within a pre-registered margin of ± 0.75 on thirty premises, without spending a token of the stream — while trailing it in connection and coherence, and trailing the reset variant of interruption. (iii) Sham pulses in random or shuffled directions displace the state more and produce far less, and a blind manipulation check finds the injected subject in the text: the direction’s content carries the effect. (iv) The surprise pattern replicates in-family at 14B. (v) Descriptively, the guarded pulse carries the flattest state trajectory in the battery, though much of the habituated carrier’s “silent orbit” is the rhythm of sentences. A verified code-search battery, reported in full, had no headroom. We map the boundaries and release the instrument, every cell, every verdict and the dated pre-registrations.

1 Introduction

Can a language model produce something genuinely new, and which interventions actually help? A sequence of studies has approached this question one substrate at a time, under a fixed measurement discipline: judged surprise, connection and coherence on windows containing only generated text (so that the judge can never be shown the intervention itself) [16], and, in verifiable domains, an exact evaluator with classical baselines [17]. At the *token* level, anti-probable sampling bought only surface novelty. At the *context* level, periodically interrupting a habituated stream with a new subject raised judged surprise from 1.8 to ~ 3.0 — the flagship positive of the program — while constant pressure (a continuity connective) was worse than nothing [16]. At the *weight* level, consolidating a model’s own verified findings moved the mean toward the known good and never beyond it, paying for mass with tails [17]; in verified search, operator packages composed additively and stalled at a construction-family plateau (companion paper, in moderation at the time of writing). Between the context and the token sits the object

none of these touched: the residual stream $h_\ell(t)$ — the thought in progress. Recent work proves that steered activations reach states that no discrete prompt reaches [15], but leaves their *value* untested; activation steering itself is an industrial control tool — removing behaviors, enforcing styles [1, 11, 18, 25, 28] — that has not been pointed at creation under a deception-resistant measure.

This paper does that. Its contributions:

1. **An instrument** (§2): a rank-1/2 operator algebra on the residual stream (§2) with model-derived directions, geometric guards, and a dimensionless dose unit; dual MLX/PyTorch backends, bit-identical on the algebra.
2. **The temporal contrast** (§3): continuous injection has no fertile band — more dose, less judged surprise, on the plain carrier and on the habituated one — while the same vectors, pulsed, climb a dose ladder; on one carrier with identical rotating directions the head-to-head pulse-minus-continuous contrast is positive but not significant at ten premises, and the two regimes fail in opposite ways (pressing degenerates; diluted pressing is inert).
3. **Equivalence in surprise, at a cost** (§4): the guarded pulse at dose 2.0 is equivalent to text interruption in judged surprise within ± 0.75 on thirty premises, trails it in connection and coherence, and trails the reset variant of interruption; sham pulses show the direction’s content, not the perturbation, carries the effect. The surprise pattern replicates in-family at 14B (§5).
4. **A spectral description** (§6): across the whole battery, with the rhythm of sentences as the null, the guarded pulse carries the flattest state trajectory; the single-cell “silent orbit” story is reduced to what survives that null.
5. **A verified null with a floor** (§7): on verified code search the same operators change no verified quantity — and the battery had no headroom, which we report as a limitation.

Every confirmatory analysis below was pre-registered (grid, primary contrast, guards, verdict tiers) and written as code before the data existed; exploratory screens are labeled as such. All deltas are paired per premise, with exact sign-flip tests (2^{10} for ten premises; Monte-Carlo sign-flip for thirty) and bootstrap 95% CIs over premises (for ten-premise batteries the percentile bootstrap under-covers — it matches a 90% t interval — so headline ten-premise contrasts also carry t intervals, and inference rests on the exact p values). Each battery declares one primary contrast, tested at $\alpha = 0.05$ without correction, and a family of secondaries reported with Benjamini–Hochberg q values over that family; a coherence guard (median paired coherence change > -1.0) accompanies every primary. The ten confirmatory premises of Ono Filho [16] are reused across this paper’s batteries (they are not held out from anything here); the equivalence extension adds twenty further premises from the same source.

2 Instrument and protocol

Operators. All operators are rank-1/2 updates from stored unit vectors, $O(d)$ per position; no $d \times d$ matrix is ever materialized. With $\hat{u}, \hat{w}$ orthonormal and $c_u = \langle h, \hat{u} \rangle$: *translation* $h + \alpha \hat{v}$; *rotor* $h + \sin \theta (c_u \hat{w} - c_w \hat{u}) + (\cos \theta - 1)(c_u \hat{u} + c_w \hat{w})$, the closed (Rodrigues) form of $\exp(\theta(\hat{w}\hat{u}^\top - \hat{u}\hat{w}^\top))$, verified against the matrix exponential to 10^{-10} ; *projection/dilation* of a subspace; *repulsion* $h + \beta (h - \mu) / \|h - \mu\|$ from an exponential moving average μ of the trajectory. Two composable *guards*: an anchor that restores the coordinate along the premise direction after the operator, and a norm shell that rescales $\|h\|$ to its pre-operator value.

Directions and dose. Concept directions are the model’s own: the unit mean hidden state of a concept text at the target layer (position 0 excluded), differenced against the premise state and normalized, $\hat{v} = (\bar{h}_{\text{concept}} - \bar{h}_{\text{premise}})/\|\cdot\|$. Doses are expressed in units of the premise-state norm at that layer, $s_\ell = \|\bar{h}_{\text{premise},\ell}\|$: a translation at dose α adds $\alpha s_\ell \hat{v}$ to every position it touches, so α is dimensionless. §5 carries α and the proportional layer (50% depth) to other models *by convention*, without per-model tuning; whether that convention is optimal there is not tested.

Terms. A *cell* is one 1500-token generation from one *premise* (opening sentence) under one *arm*. The *carrier* is the sampling regime the intervention rides on: *plain* (temperature 1.0, nucleus 0.95) or *habituated* (plain plus the count-weighted log-penalty of Ono Filho [16] over the last 512 tokens, $\ln 1.15$ per occurrence). *Text interruption* appends one of four fixed new-subject sentences to the context every 300 tokens (the injected tokens are excluded from every judged window). A *continuous* injection applies the operator at every generated position from token 301 on; a *pulse* applies it for a *burst* of 32 positions at the start of each 300-token *period* (positions $300k+1$ to $300k+32$, $k \geq 1$), with the direction rotating over the same four subject texts used by text interruption; the integrated dose of a pulse is therefore $32/300 \approx 0.107$ of the continuous dose at equal α . *Guards* are the anchor and the norm shell above. Because a cell rotates four fixed sentences once each, the verbatim self-copy that dominated the original study’s long runs (65–80% of windows) is largely designed out here (3%).

Carrier and measurement. Unless noted, generation runs on a base model (Qwen3-8B-Base, 8-bit MLX on the origin machine; bf16 PyTorch on GPU for the extension batteries, with the operator algebra verified bit-identical across backends) on the habituated carrier — the regime in which text interruption was established. Judging follows the generated-only protocol of Ono Filho [16]: a judged window is 96 generated tokens; in text-interrupted cells windows begin 32 and 160 tokens after each injection, and in every other cell (carrier and state arms, whose interventions leave no token) they sit on a 150-token grid offset by 32 (positions $150g+32$), six windows per cell evenly spread — so in state arms two of the six begin right after a burst (+32) and four sit 182 tokens after one. No window contains injected tokens or overlaps a burst; the judge also sees the 600 preceding tokens as context. The primary contrasts use all judged windows; §4 reports the sensitivity to restricting both arms to the +32 windows. The judge is Claude Opus 5 (Amazon Bedrock), $k=5$ independent calls per window, per-dimension median, on 0–10 scales of surprise, connection and coherence; its test–retest behavior was measured in the original study (91% identical medians on byte-identical cells). The judge may refuse degenerate text; refusals are counted per arm and reported wherever they are not negligible. Self-copy — a window reproducing at least half of its 12-token shingles from earlier in the stream — is flagged with the detector of the original study; in the indistinguishability battery text interruption copied 3% of its judged windows and the state arms 0%, and estimates on fresh windows only differ from the full estimates by less than 0.1. A cheap coherence proxy (mean NLL of the stream under the same model, untouched) and distinct-4 accompany every cell. Verified experiments (§7) reuse the exact bin-packing evaluator and classical baselines of Ono Filho [17].

3 The temporal contrast

Continuous injection has no fertile band. An exploratory depth×dose map (57 cells, unjudged; the cheap proxies and reading of the text; Fig. 1) located the semantic lever in the middle of the network: early-layer injection collapses generation into memorized register basins (a math-homework attractor), late-layer injection acts as token bias — at high dose the concept’s

own token fragment becomes a literal loop — and only mid-network injection transports content. The pre-registered confirmatory then swept the candidate band (layers 14/18/22, $\alpha \in [0.1, 1.5]$, continuous schedule, on the *plain* carrier, 140 cells) with judges. The primary failed in both of its pre-registered parts — the band-core contrast (layer 18, doses 0.25–0.75, the centre of the candidate band) and its coherence guard: band-core surprise minus that carrier’s baseline $-0.478 [-1.478, +0.522]$, one-sided $p = 0.81$, and the coherence guard failed (-1.08). The dose-response is monotone the wrong way: surprise $2.38 \rightarrow 0.85$ and coherence $4.83 \rightarrow 1.38$ as α rises to 1.5 (Fig. 2). More push makes text that a judge finds *less* surprising: degeneration is boring. Notably, the prompt-matched control — the same concept injected as text into this plain carrier — also failed to beat the baseline, reproducing the original finding that novelty injection needs habituation as its carrier [16].

The same vectors, pulsed, raise judged surprise. The pre-registered transposition of the interruption’s temporal structure — bursts of 32 tokens every 300, directions rotating over the same four new-subject texts used by text interruption, on the *habituated* carrier — raises judged surprise by $+0.90 [+0.45, +1.33]$ over habituation ($p = 0.0059$, coherence guard passed). Text interruption replicates in the same battery ($+1.76$, $p = 0.0029$), anchoring the instrument. One caution from the extension batteries (§4): on the PyTorch backend the same unguarded pulse gives $+0.40$ ($p = 0.13$), while the guarded pulse at dose 2.0 gives $+1.04$ ($p = 0.012$) and $+0.93$ on thirty premises ($p = 0.0002$). The robust form of the finding is the guarded pulse; the unguarded pulse at dose 1.0 is fragile across backends.

The confound, and the one-carrier test. The two results above differ in more than schedule: the continuous battery ran on the plain carrier and the pulse on the habituated one, and habituation is itself the strongest known modifier of these effects. The contrast between them is therefore suggestive, not decisive. The confirmatory that settles it puts both regimes on the habituated carrier with the *same* rotating directions and only the duty cycle varied: pulses of 32 in 300 at $\alpha = 1.0$ against continuous injection at $\alpha = 1.0$ (the per-token dose of the pulse) and at $\alpha = 0.107$ (its integrated dose), paired over the ten premises, one-sided (pulse $>$ continuous). The result is a trend, not a confirmation: pulse minus continuous $+0.33 [-0.63, +1.30]$ ($p = 0.27$) at equal per-token dose and $+0.28 [-0.12, +0.70]$ ($p = 0.13$) at equal integrated dose. What the test does establish is the asymmetry of the two regimes on the same carrier. Continuous injection at $\alpha = 1.0$ leaves surprise at the carrier’s level ($+0.07 [-0.71, +0.86]$) while collapsing coherence (2.66 against the carrier’s 4.69; clean NLL 4.4) — and, by the manipulation check below, it does carry the subject into the text (43% of windows), as degenerate prose about it. At $\alpha = 0.107$ it is inert ($+0.13 [-0.38, +0.63]$, coherence intact). The pulsed regime, meanwhile, climbs its unguarded dose ladder: $+0.40$ ($p = 0.13$), $+0.48$ ($p = 0.035$), $+0.86$ ($p = 0.043$) at $\alpha = 1.0, 1.5, 2.0$, and $+1.04$ ($p = 0.012$) guarded at 2.0. Pressing degenerates, diluted pressing does nothing, pulsing climbs; the head-to-head at equal dose is a direction we could not certify at ten premises. The plain-carrier dose-response above remains a boundary result.

4 Guards change what a dose does; equivalent in judged surprise

The unguarded pulse at dose 1.0 pays in coherence ($-1.01 [-1.85, -0.23]$ against text interruption, two-sided $p = 0.049$) while trailing it in surprise ($-0.86 [-1.76, +0.02]$, $p = 0.10$). The pre-registered guard test shows the guards do *not* repair this: at equal dose, anchor + norm shell change coherence by $+0.02 [-0.65, +0.63]$ ($p = 0.50$) while weakening the effective push (clean NLL $2.53 \rightarrow 1.83$). Under guards the ladder $\alpha = 1.0 \rightarrow 1.5 \rightarrow 2.0$ climbs monotonically in judged surprise ($2.10 \rightarrow 3.23 \rightarrow 3.40$) at roughly constant coherence (Fig. 3). Because the

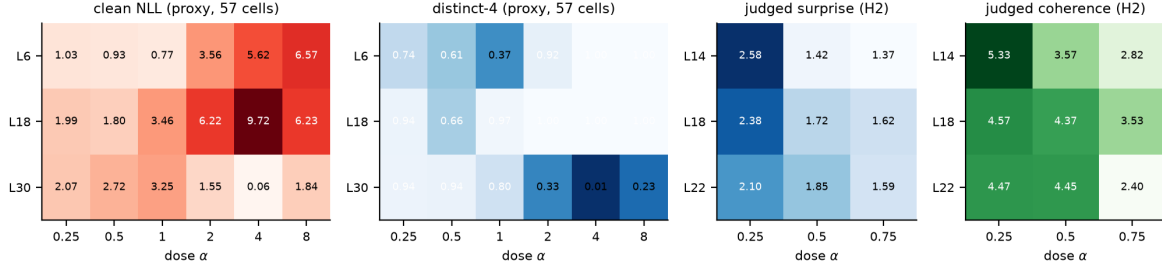


Figure 1: Depth $\times$ dose. Left pair: the exploratory 57-cell map (layers 6/18/30, continuous translation at doses 0.25–8, three premises each, no judge): clean NLL under the untouched model and distinct-4 — early-layer injection collapses into low-entropy register basins, late-layer injection at high dose loops the concept’s own token fragment. Right pair: the judged confirmatory grid (layers 14/18/22, doses 0.25–0.75, ten premises each): surprise and coherence, both falling with dose at every depth.

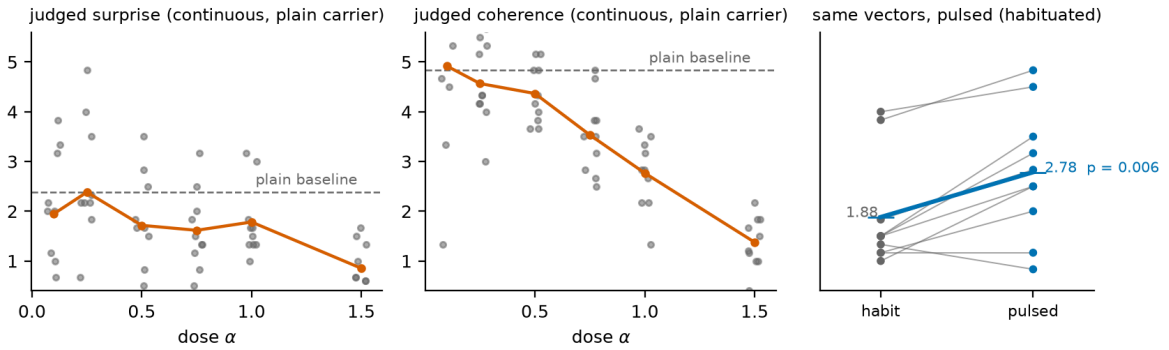


Figure 2: The temporal contrast. Left, center: judged surprise and coherence at layer 18 as continuous dose rises on the plain carrier, against that battery’s baseline (dashed) — more push, less surprise, collapsing coherence. Right: the same directions delivered as 32-token pulses every 300 on the habituated carrier, against that carrier’s own baseline ($+0.90$, $p = 0.006$). Points are premise means; the two panels differ in carrier as well as schedule (see the one-carrier test in the text).

guards act after the operator, the nominal dose overstates the displacement they leave behind; the extension battery therefore logged the *effective* displacement per burst position — the cosine between the pre- and post-operator state and the relative norm of the change (Table 1). The guarded pulse at dose 2.0 leaves the same displacement magnitude as the unguarded pulse at dose 1.0 ($\|\Delta h\|/\|h\|$ 0.75 in both) and yields $2.6\times$ its surprise gain ($+1.04$ against $+0.40$ over the carrier); the unguarded pulse at 2.0 displaces twice as much (1.53) for less ($+0.86$). So the guards do not “license dose” in the sense of admitting a larger push; they change the composition of the same push — off the premise coordinate, on the norm shell — and that composition is what the judge rewards.

At the top of that ladder sits the pre-registered comparison against the program’s strongest prompt-side intervention. Guarded pulse at dose 2.0 versus text interruption on the ten premises: surprise 3.40 vs 3.64 ($\Delta = -0.24$, 95% t interval $[-1.11, +0.62]$, two-sided $p = 0.53$), connection 2.38 vs 2.39, coherence within the pre-registered guard. A non-significant difference is not evidence of equivalence, and the paired standard deviation here (1.21) makes ten premises far too few to show it: the 90% confidence interval, $[-0.94, +0.46]$, does not fit inside any margin narrower than ± 0.95 . The pre-registered extension therefore reran the trio — habituated baseline, guarded pulse, text interruption — on thirty premises on one backend, with an equivalence margin of ± 0.75 fixed in advance (two one-sided tests; ± 0.5 reported as well). The margin is 7.5% of the 0–10 scale, about 60% of text interruption’s own effect over the carrier ($+1.22$), and about one unit of judge noise per verdict (the original study’s test–retest spread); the reader

arm	α	guards	$\cos(h_{\text{pre}}, h_{\text{post}})$	$\ \Delta h\ /\ h\ $	Δ surprise vs carrier
pulse	1.0	none	0.81	0.75	+0.40 ($p = 0.13$)
pulse	1.5	none	0.69	1.13	+0.48 ($p = 0.035$)
pulse	2.0	none	0.59	1.53	+0.86 ($p = 0.043$)
pulse	2.0	anchor + norm shell	0.72	0.75	+1.04 ($p = 0.012$)
sham, random direction	2.0	anchor + norm shell	0.56	0.94	+0.39 ($p = 0.19$)
sham, shuffled direction	2.0	anchor + norm shell	0.56	0.94	-0.09 ($p = 0.81$)
continuous	1.0	none	0.81	0.77	+0.07 ($p = 0.91$)
continuous	0.107	none	1.00	0.08	+0.13 ($p = 0.67$)

Table 1: Effective displacement and its yield. Means over fired positions and ten premises (extension battery, PyTorch bf16); surprise contrasts paired against the habituated carrier, exact two-sided sign-flip.

should note that it admits a deficit of up to 0.7. On the thirty premises the difference is -0.29 $[-0.73, +0.17]$ (two-sided $p = 0.23$), the 90% interval $[-0.68, +0.11]$ lies inside the pre-registered band (TOST $p = 0.028$; power to show ± 0.75 equivalence at the observed paired SD of 1.27: 0.87) and not inside ± 0.5 ($p = 0.18$); removing self-copied windows changes nothing (-0.29 , TOST $p = 0.029$). Both interventions replicate against the carrier on the same thirty premises (pulse $+0.93$ $[+0.49, +1.36]$, $p = 0.0002$; text $+1.22$ $[+0.80, +1.65]$, $p < 0.0001$), and the surprise contrast reproduces across backends on the original ten premises (MLX 8-bit -0.24 , PyTorch bf16 -0.29). Restricting both arms to the windows that begin 32 tokens after an intervention (two per state cell, four per interrupted cell) moves the difference to -0.06 $[-0.63, +0.52]$ (TOST ± 0.75 $p = 0.010$; ± 0.5 $p = 0.064$). The other two dimensions do not tie: the pulse trails text interruption in connection (-0.52 $[-0.96, -0.09]$, $p = 0.03$) and in coherence (-0.90 $[-1.20, -0.57]$, $p < 0.0001$) — the “identical connection” of the ten-premise battery did not survive the extension. What the claim then is: a subject placed into the model’s stream as pure activation — invisible in the text, spending no tokens, continuously dosable — produces judged surprise equivalent, within ± 0.75 on a 1–5 scale, to writing the subject down, at a measurable cost in how connected and how coherent the resulting text reads. What it is not: superiority, or evidence that the state reaches value no prompt reaches (§10).

Does the content of the pulse matter? A pulse could work by carrying a subject or merely by perturbing a habituated orbit. Two pre-registered shams keep everything about the guarded pulse — dose 2.0, schedule, guards, rotation — and replace the direction: a random unit vector per subject, and the subject’s own direction with its coordinates shuffled (same norm and entry distribution, no meaning in the model’s basis). Both shams displace the state *more* than the real pulse (Table 1) and yield far less: the real pulse beats the random-direction sham by $+0.65$ $[+0.13, +1.18]$ (one-sided $p = 0.029$, BH $q = 0.07$) and the shuffled-direction sham by $+1.13$ $[+0.65, +1.65]$ ($p = 0.001$, $q = 0.005$); against the carrier the shams themselves move $+0.39$ ($p = 0.19$) and -0.09 ($p = 0.81$). The shams are not less coherent than the real pulse (coherence $+0.23$ $[-0.38, +0.84]$ and $+0.36$ $[-0.28, +1.01]$ relative to it), so the judge is not marking them down for damage. The clean result is the shuffled sham, which keeps the direction’s norm and coordinate statistics and loses only its meaning; the random-direction sham is consistent with it. The subject’s direction, not the perturbation of an orbit, carries the effect. A blind manipulation check completes the picture: for every judged window that follows an intervention (the same $300k+32$ windows the surprise judge graded), a separate judge call names which of the four subjects, if any, the window engages, with the options shuffled per window (chance 25% among picks); text interruption is the positive control and the habituated carrier gives the base rate (Table 2). The subject reaches the judged text in about half of the text-interruption windows (49%, $p = 10^{-7}$ against chance) and in about a quarter of the guarded-pulse windows at dose 2.0 (23%; when the judge names a subject it is the injected one

arm	windows	subject named correctly	of windows with a pick	“none”
habituated carrier (base rate)	40	— (no subject)	0/6	85%
text interruption	39	49% [34, 64]	19/25	36%
guarded pulse, $\alpha=2.0$	39	23% [13, 38]	9/14	64%
guarded pulse, $\alpha=1.5$	40	25% [14, 40]	10/21	48%
guarded pulse, $\alpha=1.0$	40	5% [1, 17]	2/5	88%
unguarded pulse, $\alpha=1.0$	39	8% [3, 20]	3/7	82%
repulsion (anchored / not)	40 / 40	2% / 5%	1/4, 2/8	90% / 80%
rotor	39	3% [0, 13]	1/4	90%

Table 2: Manipulation check. A blind judge names the subject active at each judged window (chance among picks 25%); Wilson 95% CIs.

9 times in 14, $p = 0.002$), against zero on the carrier; the weak state arms — unguarded pulse, guarded pulse at dose 1.0, repulsion, rotor — transfer nothing detectable (2–8%). Content transfer by the pulse is therefore real, dose-dependent under guards, and about half as frequent as by text (in the extension battery, on the other backend: text 34%, guarded pulse 14%, reset interruption 17%, continuous injection at $\alpha = 1.0$ 43% — the subject arrives, in degenerate prose — and the two shams 3–5%, the carrier’s base rate). The pulse matches text interruption in judged surprise while placing its subject in the text less often; the shams show that what it does place there is what matters.

The comparator. Text interruption in the preserved context is the strongest prompt-side intervention of the original study *among those that keep the context*; its reset variant — the context replaced by the premise and the new subject at every period — scored as well or better in coherence there. The pre-registered extension includes that reset comparator on the ten premises, and it is the strongest arm of the battery: +1.62 over the carrier in surprise with the highest coherence (5.13). The guarded pulse trails it by -0.58 $[-0.99, -0.09]$ (two-sided $p = 0.043$, BH $q = 0.07$) with the coherence guard failing (-1.01). Against the strongest prompt-side intervention we know, then, the state channel is not at parity; it is equivalent in surprise to the preserved-context interruption and short of the reset one.

Two declared secondaries returned honest negatives. A first-version rotor (rotating the premise–subject plane, $\theta = 0.8$) underperforms text interruption (-1.53 $[-2.43, -0.63]$, $p = 0.014$, BH $q = 0.055$; a later cheap screen located its useful range at $\theta \in [1.2, 1.6]$ — an untuned operator, not a dead one). And the program’s own registered prediction failed: pulsed state repulsion *without* an anchor did not degenerate — anchored minus unanchored coherence -0.53 $[-1.18, +0.02]$, the wrong sign for the prediction — refusing the fourth appearance of the anchor law (the program’s finding, in weights and in behavior, that repulsion without a fixed reference degenerates); a stress screen then showed why — repulsion from a *moving* mean is self-limiting, because the reference chases the state. The anchor law, as established in weights and behavior, is about fixed references.

5 Generality: the pattern travels in-family

The pre-registered generality battery reran the decisive trio — habituated baseline, guarded pulse at dose 2.0, text interruption — on three untouched models, with the layer at 50% of each network’s depth and the dose carried over unchanged in premise-norm units (a convention, not a calibration).

Qwen3-14B (same family, larger): full replication. Pulse over habituation +1.083 $[+0.567, +1.617]$, $p = 0.0020$, coherence guard clean; against text interruption, an exact tie of the means: surprise 3.48 vs 3.48 ($p = 0.99$; ten premises, so again indistinguishable rather than

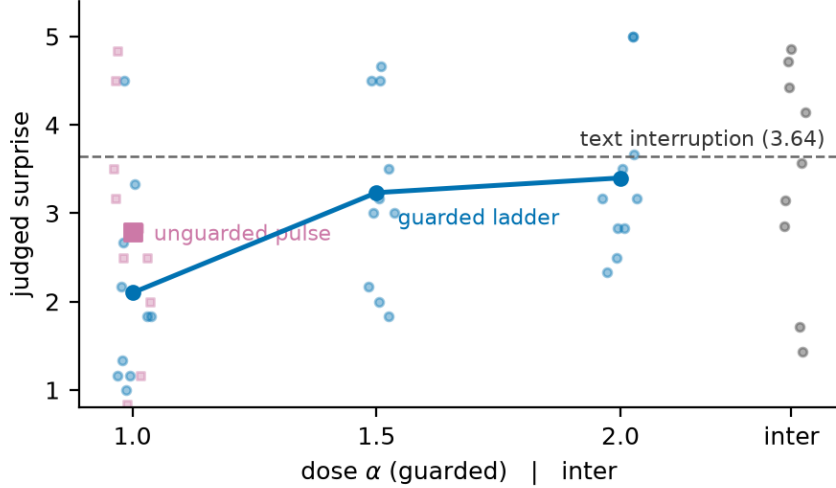


Figure 3: Guards change what a dose does. Judged surprise of the guarded pulse at doses 1.0/1.5/2.0 (points), with the unguarded pulse (square) and the text-interruption reference (dashed line), ten premises each. The ladder reaches the reference within the resolution of ten premises; the equivalence test is in the text.

model	pulse – habit (primary)	pulse – inter	verdict
Qwen3-8B (origin, 10 premises, MLX)	+0.90 [+0.45, +1.33], $p = 0.006$	−0.24 [−1.11, +0.62] ^t , $p = 0.53$	indistinguishable
Qwen3-8B (30 premises, PyTorch)	+0.93 [+0.49, +1.36], $p = 0.0002$	−0.29 [−0.73, +0.17], $p = 0.23$	equivalent (± 0.75)
Qwen3-14B	+1.08 [+0.57, +1.62], $p = 0.002$	+0.005 [−0.44, +0.43], $p = 0.99$	tie of means
Qwen3.8-27B (aligned)	+0.20 [+0.02, +0.38], $p = 0.041$	+0.14 [−0.03, +0.31], $p = 0.16$	no reading (carrier breaks)
OLMo-2-13B	−0.09 [−0.32, +0.11], $p = 0.78$	−0.09 [−0.37, +0.11], $p = 0.68$	null (register basin; text −0.00 too)

Table 3: The generality ladder. Judged-surprise contrasts, paired over the ten premises; doses in premise-norm units and layers at 50% depth, carried over by convention without retuning. “Tie” means indistinguishable at $n = 10$, not equivalence. Intervals are percentile bootstraps except the one marked t (95% t interval; the percentile interval for that contrast, [−0.94, +0.47], matches a 90% t interval).

equivalent). The dimensionless dose and proportional-depth layer worked without retuning on this model.

Qwen3.8-27B (current generation, aligned): the carrier breaks, not the phenomenon. Under our base-model protocol — which bans EOS and forces 1500 tokens — this chat-trained model opens with fluent prose and collapses into token salad past its natural turn end (judge notes: “degenerate token soup”). A nominally significant pulse effect inside that salad ($p = 0.041$) does not count as support, and we do not count it. Aligned carriers need their own protocol (chat-formatted seeding, free EOS, natural lengths); this is future work, not a footnote.

OLMo-2-13B (different family): a real null with a carrier caveat. Pulse minus habituation −0.09 ($p = 0.78$). All three arms, however, sit in an encyclopedic register basin (judged coherence 0.4–0.6 everywhere), the OLMo analogue of the register basins of §3. Judge refusals were heavy on this model and on Qwen3.8-27B — 170/214/242 (OLMo) and 208/206/190 (Qwen3.8) of the scheduled judge calls in the habituated/interrupted/pulsed arms respectively — but uniform across arms, so they cost coverage rather than bias a contrast; on the origin model they were negligible (6 in the indistinguishability battery). Whether the OLMo null reflects family-dependence of the phenomenon or miscalibration of layer and dose for this family cannot be separated here; per-model calibration is registered future work. The generality claim we make is therefore exactly this: *the pattern replicates across scale within a family; family-generality is open.*

arm	state prominence [CI]	Δ vs habit (p)	punct. prominence	$P_{\text{state}=P_{\text{punct}}}$
habit	0.039 [0.016, 0.070]	—	0.132	4/10
inter	0.016 [0.009, 0.026]	−0.023 (0.22)	0.118	3/10
pulse ($\alpha=1$)	0.035 [0.010, 0.080]	−0.004 (0.77)	0.137	4/10
pulseG ($\alpha=1$)	0.054 [0.016, 0.113]	+0.014 (0.57)	0.157	1/10
pulseG ($\alpha=1.5$)	0.013 [0.008, 0.018]	−0.027 (0.02)	0.100	3/10
pulseG ($\alpha=2$)	0.010 [0.007, 0.015]	−0.029 (0.006)	0.095	3/10
repelA	0.022 [0.012, 0.038]	−0.018 (0.12)	0.113	4/10
repelN	0.028 [0.010, 0.050]	−0.012 (0.64)	0.115	1/10
rotor	0.078 [0.013, 0.181]	+0.039 (0.91)	0.169	3/10

Table 4: Spectra of the whole battery. Dominant detrended local peak of the state autocorrelation per cell; bootstrap CIs over the ten premises; paired differences against the habituated carrier with exact sign-flip p ; the sentence-boundary autocorrelation as the rhythm null; and how often the state period equals the punctuation period within 20%.

6 Spectra: orbits, sentence rhythm, and what flattening survives

Why does the pulse work where pressure fails? A measurement study replayed finished cells through the model, capturing the mid-layer trajectory, and compared the state autocorrelation $A_{\text{state}}(\tau)$ (unit-normed, mean-centered states) against the token autocorrelation (the ground truth of literal loops), reading the strongest detrended local peak of each ($\tau \geq 5$; literal period-2 loops sit below the reader’s floor and are noted as a blind spot).

The prominence ladder in the illustrative cells is clean (Fig. 4). A literal phrase loop shows a state period matching the phrase length ($P = 7$, prominence 0.44). Register collapses — the quiz mode, the math-exercise mode — show matched token–state periodicity at their block rhythms ($P = 123$ and 84). And a habituated cell that repeats no phrase still shows matched token–state periodicity at $P \approx 30$ (prominence 0.18/0.15), whose text reads as sentence-level cycling through the same cast — the “return to the theme, not the sentence” of Ono Filho [16].

Those are single cells, and the population reading is more modest. Replaying every cell of the indistinguishability battery (90 cells, 9 arms, 10 premises each) and adding the null a state period has to beat — the autocorrelation of the sentence-boundary indicator, i.e. the rhythm of sentences — gives Table 4. Sentence rhythm is the dominant periodicity in every arm (punctuation prominence 0.10–0.17); the state prominence of the habituated carrier is small (0.039 [0.016, 0.070]) and its period coincides with the punctuation period in 4 of 10 cells — including the $P=30$ cell above, whose punctuation period is also 30. What survives as an orbit-flattening effect is specific: the guarded pulse at dose 2.0 lowers state prominence against the paired carrier by −0.029 [−0.057, −0.008] (exact two-sided $p = 0.006$), and at dose 1.5 by −0.027 ($p = 0.02$); text interruption trends the same way (−0.023, $p = 0.22$); the unguarded pulse, the rotor and moving-mean repulsion do not flatten anything. One more caution: the punctuation prominence also falls in the winning arm (0.095 against 0.132), and per-premise changes in state and punctuation prominence correlate ($r \approx 0.6$), so part of the state flattening is the sentence rhythm flattening. Masking the punctuation period ($\pm 20\%$) leaves a residual state flattening for the guarded pulse at 2.0 (−0.023 [−0.048, +0.002], exact $p = 0.010$) and at 1.5 ($p = 0.02$), not for interruption ($p = 0.08$). We therefore retract the general form of the claim: the interventions a judge rewards are not all spectral flatteners, and the habituated “silent orbit” is largely the rhythm of sentences. The specific form — the winning state arm carries the flattest trajectory in the battery, part of which survives the sentence-rhythm null — is descriptive, not causal.

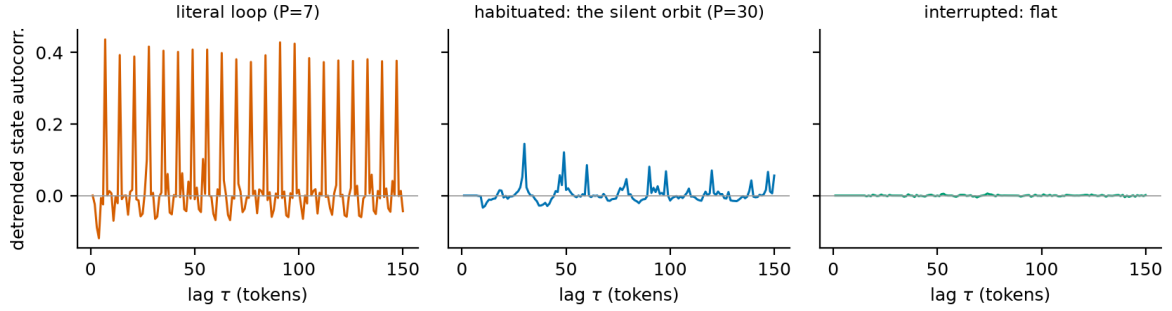


Figure 4: Silent orbits and their removal. Detrended state autocorrelation for a literal loop (state period = phrase length), the habituated carrier (the silent orbit at $P \approx 30$, matched in tokens), and an interrupted cell (flat).

arm	candidates	valid	distinct	mean gap	best gap	finds
carrier (none)	293	165	165	+0.0215	−0.00334	2
wildcard pulse	278	121	120	+0.0194	−0.00334	1
domain pulse	308	161	157	+0.0242	−0.00334	3

Table 5: Verified battery, full ledger. Gap = train excess over min(best-fit, first-fit); “find” = a program beating both classics on its variant. Variants with headroom for the carrier: 0 of 16.

7 The verified null, and its floor

Do these operators buy anything a verifier can price? The pre-registered battery ran three arms — untouched carrier, pulses toward narrative subjects (the cross-register wildcard), pulses toward the domain’s own thinking prompts — inside verified bin-packing notebook generation (16 distribution variants including the far family of Ono Filho [17], 6 notebooks each, 288 notebooks), scoring with the exact evaluator against best-fit and first-fit.

The pre-registered primary is null: per-variant best distance to the classics, wildcard minus control +0.00014 [+0.00000, +0.00036], one-sided $p = 1.0$; transfer and far-family contrasts null; the single best gap across all three arms identical to five decimals (−0.00334). The full ledger explains the identity (Table 5): the untouched carrier had already matched or beaten the classics in **16 of 16** variants, so the measure sat at its floor in every cell and no arm could improve it. The battery therefore does not support a dissociation between judged experience and verified value; it shows that the verified task, at this scale, offers no headroom for any intervention to be measured on. What the ledger does show is a cost: the wildcard arm produced fewer valid programs (121 against 165 for the carrier) at the same best value. (An earlier 120-notebook screen had produced a lone wildcard find; the confirmatory dissolved it into sampling noise — all arms find occasionally at $n=6$.) A first-version route-blocking operator — projecting out the top principal directions of a previous solution’s trajectory — behaved as a mild handicap: production down, no depth gained, and a calibration curve showing that the top trajectory components carry the code-writing machinery itself (blocking ≥ 4 directions removes the faculty, not the idea).

Consistent with the attractor results of Ono Filho [17], the verified value reachable by this carrier behaves as a constant that state stimulation did not move; whether it *could* move it on a task with headroom is untested here.

8 Boundaries

For designers of state interventions, the mapped edges: (i) *depth matters more than dose* — early layers destabilize into register basins, late layers are token bias (at high dose the injected concept’s token fragment loops literally: the late network is a mouth, not a mind); (ii) *guards change the composition of a push, not its size*; (iii) *moving-reference repulsion is self-limiting* — collapse requires a fixed reference, which reframes the anchor law; (iv) *aligned models are a different carrier* — forced continuation past the turn end manufactures degeneration that no operator caused; (v) judged-surprise instruments punish degeneration (they cannot be gamed by noise), but they refuse some degenerate text outright, which costs coverage and must be reported.

9 Related work

Activation steering. Additive steering vectors [18, 25], truthfulness intervention [11], representation engineering [28], SAE features [3, 23], function vectors [24] and refusal-direction ablation [1] steer style, safety and factual behavior; alignment training itself has been read as learning to steer around such directions [8], and industrial agents already use activation edits routinely for control [2]. Three refinements of the additive vector anticipate elements of our instrument: mean-centring corrects the direction for the dataset’s offset [7]; Householder pseudo-rotation edits direction while preserving magnitude [19], the norm-preserving intent our rotor and norm shell share (our rotor is the closed-form plane rotation rather than a reflection); and conditional activation steering [9] gates the edit on a condition — a *when* chosen by the input, where our pulse chooses it by the clock. Tan et al. [22] document how unreliable steering vectors are across prompts, which is one reason we report per-premise pairs and effective displacement rather than nominal dose. STARS [27] is closest in spirit — multiple orthogonal steering directions for diverse generation — but evaluates without a prompt-matched rival or a deception-resistant judge protocol, and uses translations only. Mishra et al. [15] prove steered states are unreachable by prompts but do not test their value; our indistinguishability-not-superiority result is a first value measurement in that opened space, and it constrains easy versions of the fertility hypothesis. Weight-space analogues — task arithmetic [6], creativity weight-steering [20] — edit the persistent model where we perturb the transient thought.

Repetition and orbits. The loop we measure as a silent orbit has a literal ancestor in the repetition problem: its self-reinforcing dynamics [26], its roots in training data [10], and dedicated neurons [4]. Habituation suppresses the literal form; the state-space rhythm that survives it is what §6 characterizes. Rotations of the residual stream are not exotic to the architecture — rotary position embeddings [21] rotate query-key pairs by position — but ours act on content directions, not positions.

Mechanistic intervention. Causal tracing and model editing record and modify internal routes to *explain* behavior [14]; our route-blocking screen inverts the direction — recording a route to *forbid* it — and reports the first calibration of how much of a trajectory can be blocked before the faculty, not the idea, is removed.

Decoding-time control. Degeneration under maximization [5] anticipates our salad regimes; contrastive decoding [12] and proxy tuning [13] intervene through logits of a second model, a surface channel our late-layer results effectively reproduce (late-layer injection $\approx$ token bias); the mid-network channel studied here is precisely what logit-level methods cannot reach.

The program. Protocol, habituation and interruption from Ono Filho [16]; consolidation, tails and the attractor from Ono Filho [17]; the operator-package factorial from the companion verified-search paper. The AAR study [2] independently reproduces the program’s telemetry (method monocultures, flat idea diversity under climbing scores, capability anchors) at industrial scale in an alignment-mitigation setting — a mass regime, complementary to the frontier questions here.

10 Discussion and limitations

What equivalence means. A new channel into the model’s ongoing thought exists, is dosable and reversible, spends no context, is invisible in the text, and — within ± 0.75 on thirty premises — produces judged surprise equivalent to the preserved-context interruption it was modeled on. It does so by carrying the subject (the shams fail; the manipulation check finds the subject in the text) and at a cost the text channel does not pay: lower connection, lower coherence, and a clear deficit against the reset interruption. What it does not mean: superiority. The strong fertility hypothesis — that prompt-unreachable states yield value prompts cannot — remains open; our continuous-injection null and the state $\approx$ prompt tie in the easy setting are its first constraining data points.

Experience versus value. We had hoped to report a dissociation — state operators changing what a judge experiences while leaving what a verifier prices untouched. The verified battery cannot carry that claim: its measure had no headroom. What survives is weaker and still worth stating: on the one verified task we ran, the operators that a judge cannot tell from the written word produced no verified gain and fewer valid programs.

Limitations. The ten confirmatory premises are reused across this paper’s batteries (each analysis pre-registered before its data, but adaptation risk accumulates and is declared; the equivalence extension adds twenty premises unused by any earlier battery of this paper). The judge is a single instrument (one model family; its test–retest behavior was measured in 16); generality is established across scale, not family; doses and layers were calibrated on the origin model and carried to others by convention (50% depth, premise-norm units) rather than per-model tuning; the verified domain is one task family and, as reported, had no headroom. The two backends do not give the same carrier: on the ten confirmatory premises the habituated 8-bit MLX carrier scores 1.88 in surprise against 2.48 for the bf16 PyTorch carrier, while the interventions land within 0.1–0.2 of each other across backends (2.78/2.87 unguarded pulse, 3.40/3.52 guarded, 3.64/3.80 text); contrasts against the carrier are therefore larger under quantization, and the unguarded pulse at dose 1.0 is significant on one backend only. The judge was the same model on both, called on different dates. Bit-identical operators do not imply comparable generation across weight precisions. The plain-carrier dose–response and the habituated pulse differ in carrier as well as schedule; the one-carrier test of §3 is the only clean temporal contrast, and at ten premises it certifies the asymmetry of failure modes, not the head-to-head difference. The equivalence result holds for judged surprise only, at the ± 0.75 margin only, and against the preserved-context interruption only. The mechanism (why 32-token pulses raise judged surprise where constant pressure degrades) is characterized spectrally, not causally.

Future work. A protocol for aligned carriers; per-model calibration for cross-family tests; the strong fertility test on a carrier where the pulse and the word are both at ceiling; and substrate-level questions — consolidating the search’s schematic memory rather than its winners, and cross-model state transfer — that the present results motivate but do not touch.

Reproducibility and provenance

The operator algebra, both backends, every generation and judging script, all cells, judge verdicts, and the dated pre-registrations are released at <https://github.com/RobertoOno/interrupting-the-loop>. All confirmatory analyses were committed as code before their data existed. Table 6 lists every battery behind a number in this paper.

battery (section)	carrier / backend	cells	judged windows	status
depth×dose map (§3)	plain / MLX 8-bit	57	0 (proxies only)	exploratory
continuous band, H2 (§3)	plain / MLX 8-bit	140	846	pre-registered
pulse, guards, H1 trio (§3–4)	habituated / MLX 8-bit	90	550	pre-registered
rotor and repulsion screens (§4)	habituated / MLX 8-bit	21	0 (proxies only)	exploratory
verified screen B-1 (§7)	code carrier / MLX	4 runs	verifier	exploratory
verified B-CONF (§7)	code carrier / MLX	288 notebooks	verifier	pre-registered
route-blocking screen (§7)	code carrier / MLX	3 runs	verifier	exploratory
spectra (§6)	replay of the 90 H1 cells	90	—	measurement
generality, Qwen3-14B (§5)	habituated / PyTorch bf16	30	190	pre-registered
generality, Qwen3.8-27B (§5)	habituated / PyTorch bf16	30	146	pre-registered
generality, OLMo-2-13B (§5)	habituated / PyTorch bf16	30	156	pre-registered
extension R4a: shams, one-carrier, reset, dose (§3–4)	habituated / PyTorch bf16	80	487	pre-registered
extension R4b: equivalence on 30 premises (§4)	habituated / PyTorch bf16	90	569	pre-registered

Table 6: Provenance. Each battery with its carrier and backend, number of cells, judged windows (each = $k=5$ judge calls), and whether it was pre-registered (confirmatory) or exploratory.

References

- [1] Andy Arditi, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction, 2024.
- [2] Yueh-Han Chen, Jiaxin Wen, and Jan Hendrik Kirchner. Automated researchers can reliably mitigate alignment failures. Anthropic Alignment Science technical report, 2026. URL <https://www-cdn.anthropic.com/7b1c44894e980876479947dcdd40716278aeffd/automated-alignment-researchers-august-2026.pdf>. Code: https://github.com/YuehHanChen/automated_alignment_researcher.
- [3] Hoagy Cunningham et al. Sparse autoencoders find highly interpretable features in language models, 2023.
- [4] Tatsuya Hiraoka and Kentaro Inui. Repetition neurons: How do language models produce repetitions? In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 483–495. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.naacl-short.41.
- [5] Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The curious case of neural text degeneration, 2019.
- [6] Gabriel Ilharco et al. Editing models with task arithmetic, 2022.
- [7] Ole Jorgensen, Dylan Cope, Nandi Schoots, and Murray Shanahan. Improving activation steering in language models with mean-centring, 2023. AACL-24 Workshop on Responsible Language Models.
- [8] Andrew Lee, Xiaoyan Bai, Itamar Pres, Martin Wattenberg, Jonathan K. Kummerfeld, and Rada Mihalcea. A mechanistic understanding of alignment algorithms: A case study on DPO and toxicity. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 26361–26378. PMLR, 2024.

- [9] Bruce W. Lee, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Erik Miehl, Pierre Dognin, Manish Nagireddy, and Amit Dhurandhar. Programming refusal with conditional activation steering. In *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.
- [10] Huayang Li, Tian Lan, Zihao Fu, Deng Cai, Lemao Liu, Nigel Collier, Taro Watanabe, and Yixuan Su. Repetition in repetition out: Towards understanding neural text degeneration from the data perspective. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023.
- [11] Kenneth Li et al. Inference-time intervention: Eliciting truthful answers from a language model, 2023.
- [12] Xiang Lisa Li et al. Contrastive decoding: Open-ended text generation as optimization, 2022.
- [13] Alisa Liu et al. Tuning language models by proxy, 2024.
- [14] Kevin Meng et al. Locating and editing factual associations in gpt, 2022.
- [15] Abhinav Mishra, Daniel Khashabi, and Tianyi Liu. Steered llm activations are non-surjective, 2026.
- [16] Roberto I. Ono Filho. Interrupting the loop: Periodic subject changes raise judged surprise and connection in base language models. arXiv:2608.19893 [cs.CL], 2026.
- [17] Roberto I. Ono Filho. Moving the mean toward the known good, not beyond it: What inference-time interventions and weight consolidation buy in open-ended generation. arXiv:2608.28886 [cs.CL], 2026.
- [18] Nina Panickssery et al. Steering llama 2 via contrastive activation addition, 2023.
- [19] Van-Cuong Pham and Thien Huu Nguyen. Householder pseudo-rotation: A novel approach to activation editing in LLMs with direction-magnitude perspective. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 13737–13751. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-main.761.
- [20] Anna Schapiro et al. Creativityneuro: Weight steering for divergent thinking in language models, 2026.
- [21] Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *Neurocomputing*, 568:127063, 2024. doi: 10.1016/j.neucom.2023.127063.
- [22] Daniel Tan, David Chanin, Angus Lynch, Brooks Paige, Dimitrios Kanoulas, Adrià Garriga-Alonso, and Robert Kirk. Analysing the generalisation and reliability of steering vectors. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024.
- [23] Adly Templeton, Tom Conerly, Jonathan Marcus, Jack Lindsey, Trenton Bricken, Brian Chen, Adam Pearce, Craig Citro, Emmanuel Ameisen, Andy Jones, Hoagy Cunningham, Nicholas L Turner, Callum McDougall, Monte MacDiarmid, C. Daniel Freeman, Theodore R. Sumers, Edward Rees, Joshua Batson, Adam Jermyn, Shan Carter, Chris Olah, and Tom Henighan. Scaling monosemanticity: Extracting interpretable features from Claude 3 Sonnet. Transformer Circuits Thread, 2024. URL <https://transformer-circuits.pub/2024/scaling-monosemanticity/index.html>.

- [24] Eric Todd et al. Function vectors in large language models, 2023.
- [25] Alexander Matt Turner et al. Steering language models with activation engineering, 2023.
- [26] Jin Xu, Xiaojiang Liu, Jianhao Yan, Deng Cai, Huayang Li, and Jian Li. Learning to break the loop: Analyzing and mitigating repetitions for neural text generation. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.
- [27] Wei Zhu et al. Stars: Inference-time stiefel activation steering for diverse generation, 2026.
- [28] Andy Zou et al. Representation engineering: A top-down approach to ai transparency, 2023.